Staff answers and checks

Starter: burning fossil fuel adds carbon dioxide and extra greenhouse gas slows heat loss (Autumn 2 Week 4). The Sun is a star, Earth orbits the Sun, or the Moon orbits Earth (Autumn 2 Week 3). Friction can slow a moving toy (Autumn 1 Week 3). Check actual taught coverage.

Model: choose toy and surface for a motion question, measure distance and test friction as one possible cause. The source notes a single run and requires repeats before a cause claim. For day and night choose torch and marked ball. The missing shared observation is the marked place moving into shade or away from light.

Sort in source order: toy slowing = toy and surface; a place in night = torch and ball; climate warming = warming cards. The diagram is simplified and its climate line is not data. Day/night involves rotation, whereas a climate pattern needs records over years.

Step check: first, a toy run with the same start. Hinge: B, second, Friction at the trolley surface. The other choices substitute orbit length, daylight or whole-Earth climate for the changed surface. Parallel hinge: a block pulled over two surfaces needs friction, not Earth’s rotation.

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Staff answers · task, limits and next move

The practical uses the same toy, start and launch method, with two measured runs on rough paper and two on smooth. Record actual results, including units. Repeat safely and note variation. Do not supply invented measurements or infer that one uncontrolled difference proves roughness caused it.

Independent core: choose two stations, give a claim and a matching clue for each, and name one limit. Supported uses a question–model match with the same core supported by an adult; Independent adds claim and clue; Aspire compares models, clues and limits. Go deeper asks what two models can show or miss. If two toy runs differ, examine start/push controls rather than claiming friction alone.

Useful limits: toy runs do not test global climate; a torch/ball model is not to scale and leaves out much of the real Earth–Sun system; a warming-chain card describes a mechanism and is not a measurement record. The climate line in the source picture must not be mistaken for data.

Review repairs a mismatched clue or adds a limit. Exit names model, claim and clue and what changed after feedback. Keep each learner’s evidence and adult read-back. Next weekly line: Consolidate and revisit key ideas through investigation. Career link: an engineering technician matches a test and evidence before changing a design.

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Staff guidance · pair, access and feedback

The pair works with a shared model and individual evidence. One explains and one asks for the clue, then they swap. Hear both independent reasons before moving on. Re-model an unsure learner's step and use a fresh parallel question while the other continues.

No cold calling or forced public reading. A learner may pass. Point, sign, speak, draw, type, write, direct an adult or use exact scribing. Read the task exactly when needed. Keep the science challenge and allow planning in any language. EAL is language, not SEN.

Use a stable tray or large cards and the agreed accessible input. Give processing time. Agree a quiet supervised pause and a brief first–then cue. Re-enter at the last completed step. Do not gather evidence while a learner is distressed.

Record actual VF, WS, I, NS(+), E, R, // and ? only when they occur. Keep deepest adult prompt separate from the pupil response. A real adult receives and names back the science. R requires the pupil's response to an NS(+) next step and real Audience; never award it for a click or completion. Keep the pupil's own wording and own-colour response.

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Staff guidance · evidence and handover

Keep each first response, result or labelled explanation, the feedback and the pupil’s edit or justified decision to keep it. E requires an observable curriculum action on [SYS]. Use SCI-GROW-A2-W05 for written, spoken, practical or photographic evidence. Record prompt level, response mode, support faded and what was independent.

At Independent agree actual adult roles and swap across the week. If one adult attends, use the actual staffing plan. Follow individual support and behaviour plans. Record access, regulation and curriculum actions against the learner’s own targets without recording a diagnosis as evidence.

If behind, preserve Choose two stations. Give a claim and matching clue for each. Name one limit. and the safety teaching. After a secure check shorten repeated oral rehearsal. Record any change, reason and coverage impact. Pass checked understanding, deepest prompt and unresolved errors to Week 6. Do not replace incomplete individual work with a group product.

Evaluation: date taught; delivered as planned; criteria met; actual adaptations and reasons; changes to the scheme and coverage; evidence banked. Source evidence mapping remains [AWARD] [QUAL]; no unit or qualification award is claimed by this lesson.

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